

Mark schemes

Q1.

(a) $25\cos 17^\circ t = 12$

*Equation to find time of flight
Correct equation*

M1 A1

$$t = \frac{12}{25\cos 17^\circ} = 0.502 \text{ s}$$

ag *Correct time from correct working*

A1

3

(b) $h = 25 \sin 17^\circ \times 0.502 - 4.9 \times 0.502^2$

*Substituting $t = 0.502$ into an expression
for the height of the ball*

M1

Correct expression

A1

$$= 2.43 \text{ m}$$

Correct height

A1

3

(c) $v_x = 25\cos 17^\circ$

*Finding horizontal and vertical
components of velocity*

M1

$$v_y = 25\sin 17^\circ - 9.8 \times 0.502 = 2.390$$

Horizontal component correct

A1

Vertical component correct

A1

$$\tan \alpha = \frac{25\cos 17^\circ}{2.390}$$

Use of tan to find angle

M1

$$\alpha = 84.3^\circ$$

Correct angle

A1

5

(d) Ball is a particle

First assumption

B1

No air resistance

Second assumption

B1

2

[13]

Q2.

(a) $150g - T = 150a$

Three term equation of motion for tank.

M1

$$T - 200g \sin 30^\circ = 200a$$

Correct equation

A1

$$150g - 200a - 200g \sin 30^\circ = 150a$$

Three term equation of motion for trolley

M1

Correct equation

A1

$$a = \frac{150g - 200g \sin 30^\circ}{350} = 1.4 \text{ ms}^{-2}$$

Solving for a

m1

Correct equation for a

A1

ag *Correct a from correct working.*

A1

7

(b) $T = mg$

Equations of motion for both bodies.

M1

$$T = 200 g \sin 30^\circ$$

Correct equation for tank.

A1

Correct equation for trolley

A1

$$m = 200 \sin 30^\circ = 100 \text{ kg}$$

Correct mass

A1

4

[11]

Q3.

(a) $8 = 10 \cos 60^\circ t$

Horizontal equation based on travelling 8 metres

M1

Correct equation

A1

$$t = \frac{8}{10 \cos 60^\circ} = 1.6 \text{ s}$$

ag *Correct time from correct working*

A1

3

(b) $h = 10 \sin 60^\circ \times 1.6 - 4.9 \times 1.6^2$

Expression for height using $t = 1.6$

M1

Correct expression

A1

Correct height

A1

3

(c) $V = 10 \cos 60^\circ = 5$

Horizontal component of the velocity

B1

$$V = 10 \sin 60^\circ - 9.8 \times 1.6 = -7.020$$

Expression for vertical component of velocity

M1

Correct vertical component

A1

Finding the magnitude

M1

$$v = \sqrt{5^2 + 7.020^2} = 8.62 \text{ ms}^{-1}$$

Correct magnitude

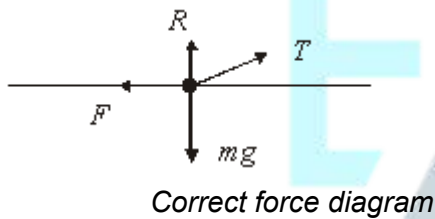
A1

5

[11]

Q4.

(a)



B1

1

(b) $R + T \sin 40^\circ = 50 \times 9.8$

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$$R = 490 - T \sin 40^\circ$$

Three term equation of motion

M1

Correct equation

A1

Correct expression for R

Use of $F R = \mu$

A1

3

(c) $F = 0.6 (490 - T \sin 40^\circ)$

Use of $F = \mu R$

M1

$$= 294 - 0.6 T \sin 40^\circ$$

ag Correct result from correct working

A1

2

(d) $T \cos 40^\circ - (294 - 0.6 T \sin 40^\circ) = 50 \times 0.5$

$$T = \frac{319}{\cos 40^\circ + 0.6 \sin 40^\circ} = 277 \text{ N}$$

Four term equation of motion

M1

Correct equation

A1

Solving for T

M1

Correct T

A1

4

[10]

Q5.

(a) $14a = 14g \sin 45^\circ - T$

Equation of motion for one particle

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M1

$$6a = T - 6g$$

Correct equation

A1

$$14a = 14g \sin 45^\circ - (6a + 6g)$$

Equation of motion for other particle

Correct equation

M1 A1

$$a = \frac{14g \sin 45^\circ - 6g}{20} = 1.91 \text{ ms}^{-2}$$

Correct a from correct working

M1 A1

(b) $T = mg$

Equation for one particle

M1

$T = 14g \cos 45^\circ$

*Equation of other particle
correct m*

M1 A1

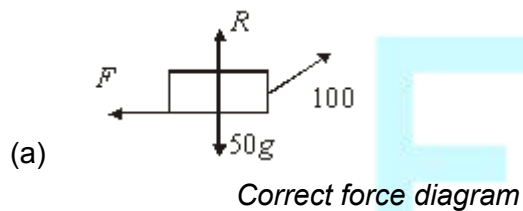
$m = 14 \cos 45^\circ = 9.90 \text{ kg}$

A1

4

[10]

Q6.



B1

1

(b) $50 \times 9.8 = R + 100 \sin 30^\circ$

*Resolving vertically
Correct equation*

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©2025 Exam Papers Practice. All rights reserved. M1 A1

$R = 440 \text{ N}$

Correct F from correct working

A1

3

(c) $100 \cos 30^\circ \leq \mu \times 440$

Use of $F \leq \mu R$ or $F = \mu R$

M1

$\therefore k = 0.197$

A1

$\mu \geq 0.197$

Correct equation
 Correct k from correct working

A1

(d) $50a = 100 \cos 30^\circ - 0.1 \times 440$

Resolving horizontally to obtain a 3 term equation of motion

M1

3

Correct equation

A1

Solving for a

m1

$a = 0.852 \text{ ms}^{-2}$

Correct a

Allow 0.680 or 0.681

A1

4

[11]

Q7.

(a) $0 = 10 \sin 70^\circ t - 4.9t^2$

Equation for height equal to zero

Solving for t

M1 M1

$t = 0 \text{ or } t = \frac{10 \sin 70^\circ}{4.9} = 1.918 \text{ s}$

Correct t

Calculating range

A1 M1

$R = 10 \cos 70^\circ \times 1.918 = 6.56 \text{ m}$

Correct range

A1

5

(b) $-2 = 10 \sin 70^\circ - 4.9t^2$

*Forming equation for vertical motion
 when ball lands*

LHS correct

RHS correct

M1 A1 A1

$$4.9t^2 - 10 \sin 70^\circ t - 2 = 0$$

$$t = 2.11 \text{ or } -0.193$$

Solving quadratic equation

Correct solution

M1A1

$$R = 10 \cos 70^\circ \times 2.11 = 7.22\text{m}$$

Calculating range

Correct range

M1A1

7

[12]

Q8.

(a) $R + 100 \sin 20^\circ + 60 \sin 50^\circ = 490$

Resolving vertically

Correct equation

M1 A1

$$R = 410$$

A1

3

(b) (i) $100 \cos 20^\circ - 60 \cos 50^\circ - F = 25$

Four term equation of motion

Correct equation

M1 A1

$$F = 30.4$$

Solving for F

Correct F from from correct working

M1 A1

4

(ii) $30.40 = \mu \times 409.8$

Use of $F = \mu R$

M1

$$\mu = 0.0742$$

Correct μ

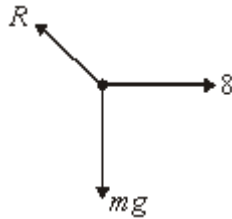
A1

2

[9]

Q9.

(a)



Correct force diagram with labels

B1

1

(b) $R \cos 30^\circ = 8$

Resolving horizontally to get two terms

Correct equation

M1

A1

$$R = \frac{8}{\cos 30^\circ} = 9.24$$

ag *Correct answer from correct working
(other methods: M1 A1 if correct)*

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A1

3

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(c) $R \cos 60 = 9.8m$

*Resolving horizontally to get two terms,
with 8 not included*

M1

Correct equation

A1

$$m = \frac{8 \cos 60^\circ}{9.8 \cos 30^\circ} = 0.47$$

Correct m to 2 sf

A1

3

(Resolving perpendicular to the plane: M1A1)

for equation and A1 for final answer)

[7]

Q10.

(a) $5 = 32\sin 60^\circ t - 4.9t^2$

M1

Equation for vertical motion with ± 5

M1

LHS correct

A1

RHS correct

A1

$$4.9t^2 - 32\sin 60^\circ t + 5 = 0$$

$$t = 0.1866 \text{ or } 5.4691$$

Solving quadratic

m1

5.47 seconds

Selecting larger answer from two solutions or obtaining one answer with a reason

A1

5

(b) $32\cos 60^\circ \times 5.469 = 87.5 \text{ m}$

Equation for horizontal motion

M1

Correct range

A1

2

(c) $v_H = 32\cos 60^\circ$

Horiz. component of velocity seen or used

B1

$$v_V = 32\sin 60^\circ - 9.8 \times 5.469 = -25.88$$

Finding vertical component of velocity

M1

Correct vertical component

A1

$$v = \sqrt{16^2 + 25.88^2} = 30.4 \text{ ms}^{-1}$$

Finding magnitude

m1

Correct speed

(Note Max Height = 39.2 m from $t = 2.83$)

A1

5

[12]

Q11.

(a) $R = 60 \times 9.8 \times \cos 40^\circ = 450 \text{ N}$

Resolving perpendicular to the plane

M1

Correct reaction force

A1

2

(b) $F = 60 \times 9.8 \times \cos 50^\circ = 378$

Resolving parallel to the plane

M1

Correct F

A1

$$F \leq \mu R$$

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Use of $F \leq \mu R$ or $F = \mu R$

m1

$$\mu \geq 0.839$$

Correct inequality

Accept 0.840

A1

4

(c) $F = 0.2 \times 60 \times 9.8 \cos 40^\circ$

$$60a = 60 \times 9.8 \cos 50^\circ - 0.2 \times 60 \times 9.8 \cos 40^\circ$$

Use of $F = \mu R$ to find F but not with $R = 60g$

M1

Three term equation of motion

M1

Correct equation

A1

$$a = 4.80 \text{ ms}^{-2}$$

Correct acceleration

A1

5

[11]

Q12.

(a) (i) $20 \sin 30^\circ t - 4.9t^2 = 0$

Forming equation for time of flight

M1

Correct equation

A1

$$t = 0 \text{ or } t = \frac{20 \sin 30^\circ}{4.9} = 2.04 \text{ (to 3 sf)}$$

Solving quadratic equation

M1

ag *Correct solution from correct working*

A1

4

(ii) $R = 20 \cos 30^\circ \times 2.04 = 35.3 \text{ m}$

Calculation of range

M1

Correct range

A1

2

(b) $20 \sin 30^\circ t - 4.9t^2 = 2$

Equation to find t at height of 2

M1

Correct equation

A1

$$4.9t^2 - 10t + 2 = 0$$

$$t = 0.2248 \text{ or } t = 1.82$$

Correct times

A1

$$v_y = \sin 30^\circ - 9.8 \times 0.2248 = 7.797$$

Calculating vertical component

M1

Correct vertical component

A1

$$v_x = 20 \cos 30^\circ = 17.32$$

Finding horizontal component

B1

Finding speed from components

M1

$$v = \sqrt{v_x^2 + v_y^2} = 19.0 \text{ ms}^{-1} \text{ (to 3 sf)}$$

Correct speed

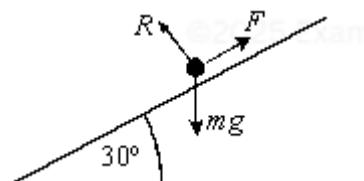
A1

8

[14]

Q13.

(a)



Correct force diagram

B1

1

(b) $R = 5 \times 9.8 \cos 30^\circ$

Finding normal reaction

M1

$$F = 0.2 + 5 \times 9.8 \sin 30^\circ = 8.5 \text{ N (to sf)}$$

Finding friction

M1

Correct friction

A1

3

(c) $5 \times 9.8 \sin 30^\circ - 0.2 \times 5 \times 9.8 \cos 30^\circ = 5a$

Three term equation of motion

M1

Correct equation

A1

$$a = \frac{5 \times 9.8 \sin 30^\circ - 0.2 \times 5 \times 9.8 \cos 30^\circ}{5}$$

Solving for a

M1

$$= 3.20 \text{ ms}^{-2}$$

Correct a

A1

4

(d) $v^2 = 0^2 + 2 \times 3.203 \times 1.2$

Constant acceleration equation with $u = 0$

M1

Correct equation

A1 ft.

$$v = 2.77 \text{ ms}^{-1}$$

A1 ft.

3

[11]

Q14.

(a) $0 = V \sin \alpha - gt$

Equation for t at max. height

M1

$$t = \frac{V \sin \alpha}{g}$$

Solving for t

M1

Correct t

A1

$$H = V \sin \alpha \times \frac{V \sin \alpha}{g} - \frac{1}{2} \times g \times \left(\frac{V \sin \alpha}{g} \right)^2$$

Substitution into equation for height

M1

Correct expression

A1

$$= \frac{V^2 \sin^2 \alpha}{2g}$$

Correct final result from correct working

A1

6

(b) (i) $6 = \frac{V^2}{2g} \times \frac{3}{4}$

Substitution into given formula

M1

$$V = \sqrt{16g} = 12.5 \text{ ms}^{-1}$$

Solving for V

M1

Correct value

A1

3

(ii) $t = \frac{2\sqrt{16g} \sin 60^\circ}{g} = 2.213$

Finding time of flight

M1

Correct time of flight

A1

$$R = \sqrt{16g} \cos 60^\circ \times 2.213 = 13.9 \text{ m}$$

Finding range

M1

Correct range

A1

4

[13]

Q15.

(a) $15^2 = 20^2 + 500a$

Use of constant acceleration equation

M1

Correct equation

A1

$a = -0.35 \text{ ms}^{-2}$

Correct acceleration from correct working

A1

3

(b) $P - 1200 \times 9.8 \cos 86^\circ = 1200 \times (-0.35)$

Three term equation of motion

M1

Correct equation

A1

Solving for P

m1

$P = 400 \text{ N}$

Correct P

A1

4

(c) $P = 400 + 300 = 700 \text{ N}$

Including resistance force

M1

Correct value

A1

2

- (d) Expect air resistance force to decrease as car moves up the slope as it is slowing down

Air resistance

Describing variation

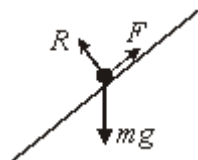
B1

2

[11]

Q16.

(a)



Correct force diagram
No extra forces

B1

1

(b) (i) $R = 80 \times 9.8 \cos 30^\circ = 679.0 \text{ N}$

Resolving perpendicular to the slope
Correct Reaction force

M1 A1

2

(ii) $F = 0.3R = 204 \text{ N}$

Multiply by 0.3
ag *Correct answer obtained*

M1 A1

2

(c) $80a = 80 \times 9.8 \sin 30^\circ - 203.7$

Three term equation of motion
Correct equation

M1 A1

$a = 2.35 \text{ ms}^{-2}$

Solving for a
awrt 2.35

m1 A1

4

[9]

Q17.

(a) $T \cos 42^\circ = 60 \cos 40^\circ$

Resolving horizontally

M1

Correct equation

A1

$$T = \frac{60 \cos 40^\circ}{\cos 42^\circ} = 61.8 \text{ N (to 3sf)}$$

awrt 61.8

A1

3

(b) $T \sin 42^\circ + 60 \sin 40^\circ = 9.8m$

Resolving vertically with three terms

M1

Correct equation with T from above

A1

$$m = \frac{T \sin 42^\circ + 60 \sin 40^\circ}{9.8} = 8.16 \text{ kg (to 3sf)}$$

Solving for m

awrt 8.16

m1

A1

4

[7]

Q18.

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(a) $0 = V \sin \alpha t - gt^2$

Equation for height equal to zero

B1

$$t = 0 \text{ or } t = \frac{2V \sin \alpha}{g}$$

Finding t

Correct time of flight

M1 A1

$$\text{Range} = V \cos \alpha \times \frac{2V \sin \alpha}{g}$$

Expression for horizontal displacement

Substituting the time of flight

M1 M1

$$= \frac{v^2 \sin 2\alpha}{g}$$

Correct final answer from correct working

A1

6

- (b) Because the maximum value of $\sin 2\alpha$ is 1.

Reason that uses maximum of sin being 1

B1

$$\alpha = 45^\circ$$

$$45^\circ$$

B1

2

(c) (i) $\frac{v^2}{g} = 35$

Substituting 35

M1

$$v = \sqrt{35g} = 18.5 \text{ m s}^{-1}$$

cao

A1

2

(ii) $R = \frac{18.5^2 \sin 60^\circ}{g} = 30 \text{ m}$

Substituting $\alpha = 30^\circ$

30 m

M1 A1

2

(or $R = 35 \sin 60^\circ = 30$)

[12]

Q19.

(a) $5^2 = 1^2 + 2 \times 4a$

Using constant acceleration equation to find a

M1 A1

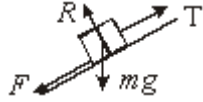
$a = 3 \text{ ms}^{-2}$

Correct equation

A1

3

(b) (i)



Correct force diagram

B1

1

(ii) $R = 20 \times 9.8 \cos 30^\circ = 170 \text{ N}$

Finding the normal reaction

M1

$F = 0.4 \times 20 \times 9.8 \cos 30^\circ = 67.9 \text{ N}$

Using $F = \mu R$

M1 A1

3

(iii) $T - 67.9 - 20 \times 9.8 \sin 30^\circ = 20 \times 3$

Four term equation of motion

Correct equation

M1 A1

$T = 226 \text{ N}$

Solving for T

Correct T

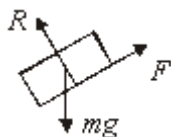
m1 A1

4

[11]

Q20.

(a)



Correct force diagram

B1

1

- (b) $R = 7 \times 9.8 \cos 30^\circ = 59.4 \text{ N}$
Resolving perpendicular to plane
Correct force

M1/A1

2

- (c) $F = 7 \times 9.8 \sin 30^\circ = 34.3 \text{ N}$
Resolving parallel to plane
Correct force

M1/A1

2

- (d) $34.3 \leq \mu \times 59.4$
Using $F = \mu R$ or $F \leq \mu R$

M1

$$\mu \geq 0.577$$

Correct inequality

A1

2

- (e) Remains at rest, as mass appears in both F and R , so inequality unchanged.
Conclusion

B1

Reason

B1

2

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[9]

Q21.

- (a) $-25 = 40 \sin 5^\circ t - 4.9t^2$

M1/A1/A1

$$4.9t^2 - 40 \sin 5^\circ t - 25 = 0$$

Equation for vertical motion

LHS correct

RHS correct

$$t = \frac{40 \sin 5^\circ \pm \sqrt{(40 \sin 5^\circ)^2 - 4 \times 4.9 \times (-25)}}{2 \times 4.9}$$

Solving quadratic equation

Correct method

m1/A1

$$= 2.642 \text{ or } -1.931$$

Hits water after 2.64 seconds

Selecting 2.64

A1

6

(b) $x = 40 \cos 5^\circ \times 2.642 = 105 \text{ m}$

Equation for horizontal motion

Correct range

M1/A1

2

(c) $v_x = 40 \cos 5^\circ$

M1/M1/A1

$$v_y = 40 \sin 5^\circ - 9.8 \times 2.642$$

Horizontal component at impact

Vertical component at impact

Vertical component correct

$$v = \sqrt{(40 \cos 5^\circ)^2 + (40 \sin 5^\circ - 9.8 \times 2.642)^2}$$

Finding magnitude

M1

$$= 45.7 \text{ ms}^{-1}$$

Correct final answer

A1

5

[13]